

Online feature Selection using Pearson Correlation Technique

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Abstract- The large volume of data that is generated nowadays in the era of real-time data processing takes a long time. Streaming feature selection is one way to get out of this real time processing challenge. In stream-wise feature selection, new features are assessed one by one for insertion into a forecasting model. When there is a large pool of acceptable characteristics, stream-wise feature selection performs well over traditional feature selection approaches, in which the features are known in advance. Over fitting can be avoided by adjusting the feature addition threshold dynamically. Unlike classic forward feature selection methods like stepwise regression, which assesses all available features at each step and choose the best one, stream-wise feature selection considers each feature only once when it is formed. In our work, we have used Pearson correlation technique for feature selection of streaming data blocks and compared with complete feature list of streaming data blocks for human activity recognition (HAR) dataset. The proposed algorithm gives promising results in terms of accuracy 94.85(%) over complete feature list.

Keywords – Data streams, feature selection, Pearson correlation

I. INTRODUCTION

A significant number of potentially predictive features are calculated by many predictive modeling methods using a given set of samples. a small portion of these traits are thought to be useful in prediction models. Pairwise interactions and data changes to the initial set of features are typically needed for superior statistical results [11]. In a prediction model only some features are vital rather than using the entire features which will be irrelevant and computationally expensive. In processing data streams where data arrives in sequential order and high speed and a lot of changes in distribution of data, it is very challenging to identify the relevant features for that particular block of data streams. We can address candidate selected features of indeterminate or even unbounded size by treating the candidate feature set as a dynamically produced stream as not every potential feature needs to be generated and confirmed.

In various circumstances, the ability to select among a set of features of varying sizes is critical. In contrast to the case where all candidate features are known in advance, the study explores a case where the candidate feature set size is unknown, or possibly infinite. The candidate features in this challenge are dynamically generated and shown one at a time with a fixed quantity of data points. Figure 1 illustrates this situation, which is referred to as streaming feature selection, and it is helpful in a number of circumstances [12].

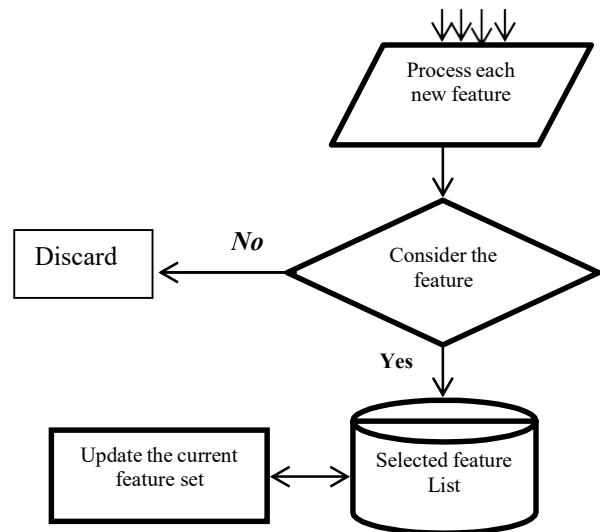


Figure 1: General feature selection framework

II. RELATED WORK

Finding the appropriate subset of features has long been thought of as the key to solving the big dimensionality problem in feature selection. Theoretically, three types of feature selection techniques are available: filter, wrapper, and embedding approaches. Utilizing evaluation criteria including distance, information, dependence, and consistency, the filter technique [1] selects features before using them to build subsets while using a classifier of choice to evaluate each classifier.

It is independent of any classifiers. The wrapper technique [2] does a heuristic search in the space of all potential feature prospective feature subset. The embedded approach [4] tries to increase classification performance while minimizing the total number of characteristics used at once. For the purpose of selecting streaming features, A grafting algorithm that is based on the stage-wise gradient descent method was proposed by Perkins and Theiler [7]. On the other hand, Grafting necessitates the knowledge of all candidate features in advance to identify the value of the tuning parameter by cross-validation before learning. Grafting is thus a quasi-streaming approach to feature selection. For the purpose of choosing a streaming feature, Information investment and alpha investing are two methods introduced by Zhou et al. [12] that use stream wise regression.. A linear regression is used to assess the enhanced model in alpha-investing, and a p-value is utilized to determine whether or not a new feature should be included to the learning model. an unique take on the Balanced Winnow algorithm [16] that performs remarkably well in single-pass NLP tasks, outperforming SVM in terms of output. Averaging (sometimes known as voting) was tested on a group of online students and found to considerably improve performance on non-NLP activities. For Winnow, Passive-Aggressive, ROMMA, and other mistake-driven online learners, averaging techniques have not been employed to examine the Perceptron method in the past. J Wang [20] proposed an online feature selection algorithm that the classifier is allowed to access the complete set of features during training and identify relevant features and also a classifier is allowed to use a subset of features to identify active features by studying sparsity regularization and truncation techniques. H. Huang [21] recommended using an unsupervised feature selection technique to deal with high-dimensional data points that arrive in streaming mode. Using matrix sketching techniques, the approach creates a continuous low-rank approximation of the input, which is then employed in a regularized regression framework to produce individual feature weights. The method simply needs one pass through the data, utilizes little storage, and executes almost instantly. Hamoodi et al. [22] proposed the Micro Cluster-Nearest Neighbor model that uses adaptive statistical summaries of absorbed data stream items. Concept drift can cause these summaries to break into new micro-clusters, alter size and position, or be eliminated (Micro-cluster death). Vadim Borisov [23] proposed CancelOut, a deep neural network layer that identifies significant streaming or static input elements. The goal is to change CancelOut's weights (W) throughout training so that "noisy" or less critical qualities are canceled out. Otherwise, only the finest characteristics that contribute to learning will be rewarded.

A. Pearson Correlation

A well-known metric for determining how comparable two features are is correlation. If there is linear dependence between two attributes, the coefficient value is 1. When there is no correlation, the coefficient value is 0. The correlation approach is employed to determine the link between two features. The linear correlation coefficient is the most well-known correlation factor. The linear Pearson correlation 'r' for a pair of variables (X, Y) is given by

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \rightarrow (\text{Equation.1})$$

r = correlation co efficient

x_i = values of each feature

y_i = values of target variable

$\bar{x}$ = mean of the features

$\bar{y}$ = mean of the target variable

B. Formal Approach of Pearson Correlation

Let $A = \{a_1, a_2, a_3, \dots, a_n\}$ be the set of N number of features and $C = \{t_1, t_2, t_3, \dots, t_n\}$ be the target variables and $V = \{v_1, v_2, v_3, \dots, v_n\}$ for the set of samples for the given data. Identify the best relevant feature $(a_i, v_i) >$ threshold for the given sample.

III. Methodology:

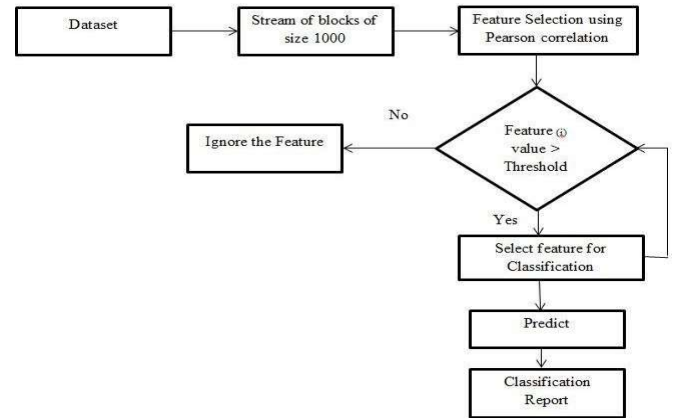


Figure 2: Framework for feature selection using Pearson Correlation

In the above figure 2, the human activity recognition (HAR) data is read continuously in block-wise stream fashion, and each block of data is subjected to the Pearson feature selection technique where each feature of that particular block with target variable correlation is found compared with the threshold value. The feature whose correlation is greater than the threshold will be added to the dictionary and further used for classification. The

procedure is reported for each block unless no more blocks remain in the dataset.

A. Algorithm

Input: $F = \{a_1, a_2, a_3, \dots, a_n\}$ be the function that defines list of features

$B = \{b_1, b_2, b_3, \dots, b_n\}$ be the data blocks from the given data samples

Output: Accuracy of tree model with feature selection

Initialization:

N = Total number of features

Batch size = 500

Threshold = 0.4

repeat

 Read the data blocks of defined batch size

for each data block **do**

 Pearson_Correlation (data_block, target)

 Build Decision tree model with feature

list

 Measure the accuracy of the data block

 with selected features

Until (no more data blocks in data source)

Calculate the correlation

Pearson_Correlation (data_block, target)

x = feature list

y = target variable

for each attribute a_i in the data_block **do**

 find the correlation using the below equation

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

if $r > \text{threshold}$ **then**

 Select the feature and add to the feature

list

 return (feature list)

else

 Discard the attribute

IV. RESULTS AND DISCUSSIONS:

A. Dataset Description:

The experiment was conducted on human activity recognition (HAR) dataset to find the most prominent streaming features using Pearson correlation. The dataset contains the binary version of human activity recognition like "walking" as our positive class with label 1 and negative class as "non-walking" with class label 0. The dataset considered for the experimentation is from standard UCI repositories. The dataset contains 562 attributes and 2 target label values and over 4500 instances.

Here are the details for each instance in the dataset:

- A body's estimated acceleration as well as the overall acceleration.
- Angular speed as measured by the gyroscope.
- With time and frequency domain variables, feature vector.

- Class label.
- one who conducted the experiment.

Decision tree classifier is implemented to measure the performance of the framework. Comparison of the complete feature list is done with Pearson correlated selected feature list using decision tree model. In comparison the Pearson correlated feature selected list performs well over complete streaming feature list in accuracy.

B. Evaluation metrics:

TABLE I: Confusion matrix

		PREDICTED VALUES	
		→	
Actual Values ↓	0	0	1
	0	TN	FN
	1	FP	TP

The above table I represents confusion matrix which is used as an evaluation metric for measuring the performance.

TP = True Positive, FP = False Positive,
TN = True Negative, FN = False Negative

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

$$\text{Average Accuracy} = \frac{\text{Accuracy of block } i}{\text{Total number of data blocks}}$$

C. Hyper parameter tuning of threshold

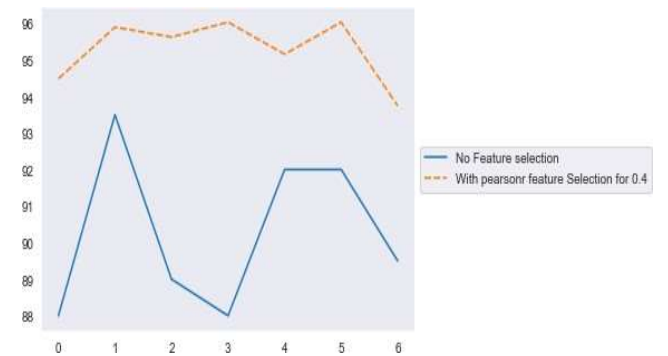


Figure 3: Accuracy comparison of feature selection v/s without feature selection for threshold 0.4

The above figure 3 demonstrates the accuracy comparison of feature selection with accuracy of complete feature list

for different data blocks. The data blocks are represented by the x-axis, while the classifier's percentage accuracy is shown on the y-axis. In comparison the accuracy with feature selection for the given threshold 0.4 performs well for different data blocks over complete feature list.

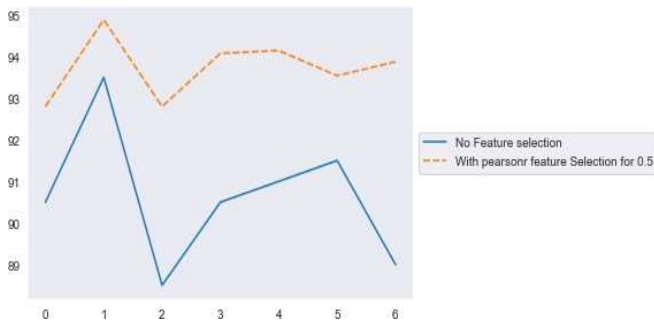


Figure 4: Accuracy comparison of feature selection v/s without feature selection for threshold 0.5

The above figure 4 demonstrates the accuracy comparison of feature selection over complete feature list of different data blocks. The y-axis displays the decision classifier accuracy in percentage, and the x-axis shows data blocks of size 500 samples. In comparison the accuracy with feature selection for the given threshold 0.5 performs well for different data blocks over complete feature list.

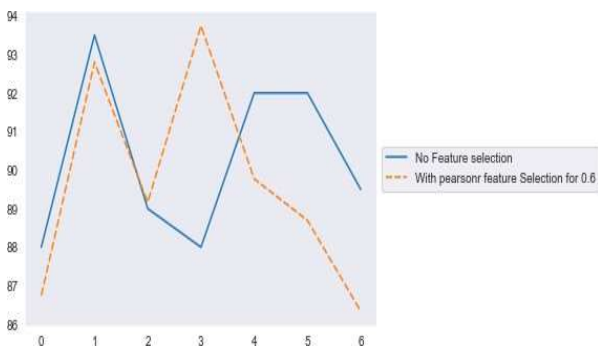


Figure 5: Accuracy comparison of feature selection v/s without feature selection for threshold 0.6

The above figure 5 demonstrates the accuracy comparison of feature selection with accuracy of complete feature list for different data blocks. The data blocks are represented by the x-axis, while the classifier's percentage accuracy is shown on the y-axis. In comparison the accuracy of data blocks 1,4,5,6 performs well with complete feature list over Pearson feature selection.

TABLE II : COUNT OF FEATURE LIST WITH ACCURACY FOR DIFFERENT THRESHOLDS AND DATA BLOCKS

Block Size	Threshold	Count of Features Selected	Accuracy with Feature Selection (%)	Accuracy without Feature Selection (%)
0-1000	0.4	215	94	88
0-1000	0.5	114	92	89
0-1000	0.6	17	86	88
1000-1500	0.4	239	95	93
1000-1500	0.5	172	94	91
1000-1500	0.6	53	92	93
1500-2000	0.4	244	95	89
1500-2000	0.5	138	93	88
1500-2000	0.6	40	89	88
2000-2500	0.4	289	96	88
2000-2500	0.5	191	94	90
2000-2500	0.6	103	93	88
2500-3000	0.4	252	95	92
2500-3000	0.5	167	93	91
2500-3000	0.6	55	89	92
3000-3500	0.4	223	96	92
3000-3500	0.5	158	93	92
3000-3500	0.6	44	88	92
3500-4000	0.4	228	93	89
3500-4000	0.5	128	94	88
3500-4000	0.6	22	86	89

0-1000	0.4	215	94	88
	0.5	114	92	89
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	0.6	53	92	93
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	0.5	138	93	88
	0.6	40	89	88
2000-2500	0.4	289	96	88
	0.5	191	94	90
	0.6	103	93	88
2500-3000	0.4	252	95	92
	0.5	167	93	91
	0.6	55	89	92
3000-3500	0.4	223	96	92
	0.5	158	93	92
	0.6	44	88	92
3500-4000	0.4	228	93	89
	0.5	128	94	88
	0.6	22	86	89

The table II gives the details about accuracy and count of selected features of streaming data blocks for HAR dataset with specified threshold $\{0.4, 0.5, \text{ and } 0.6\}$.

TABLE III: AVERAGE ACCURACY COMPARISON FOR DIFFERENT THRESHOLDS OF HAR DATASET

Block Size	Average Accuracy with feature selection for threshold 0.4	Average Accuracy with feature selection for threshold 0.5	Average Accuracy with feature selection for threshold 0.6
1-4000	94.85	93.28	89.00

Table III describes the average accuracy comparison for threshold values 0.4, 0.5 and 0.6. In comparison the average accuracy for threshold 0.4 gives the better accuracy over other threshold values.

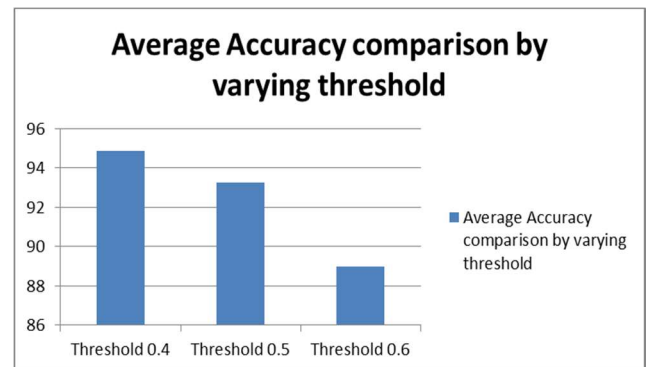


Figure 6: Comparison of Average accuracy for different threshold values

The above table and figure depicts the average accuracy comparison of different threshold $\in \{0.4, 0.5, \text{ and } 0.6\}$ values for HAR dataset. In comparison the average accuracy of feature selection using decision tree model for threshold 0.4 outperforms over other threshold values.

TABLE IV: AVERAGE ACCURACY COMPARISON OF FEATURE SELECTION V/S WITHOUT FEATURE SELECTION

Block Size	Average Accuracy with feature selection for threshold 0.4	Average Accuracy without feature selection
1-4000	94.85	90.14

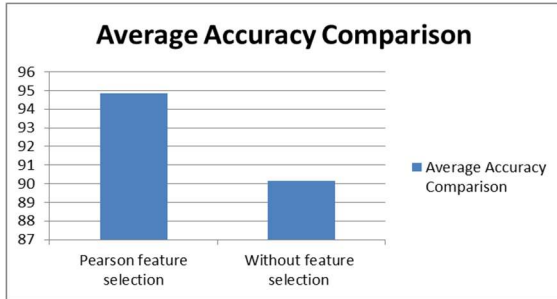


Figure 7: Average accuracy comparison of Pearson feature selection with complete feature list

The table IV and figure 7 depicts the average accuracy comparison of with and without feature selection for streaming data blocks of HAR dataset. In comparison the streaming feature selection outperforms well over without feature selection in terms of average accuracy.

V. COMPARISON OF PROPOSED METHOD WITH EXISTING METHODS

The proposed online feature selection method using Pearson correlation method is compared with top 25 percent of existing online feature selection methods. In comparison the proposed method provides the satisfactory results in terms of accuracy and time.

TABLE V: AVERAGE ACCURACY AND TIME COMPARISON OF FEATURE SELECTION V/S ONLINE FEATURE SELECTION METHODS

Method	OFS [20]	FSDS [21]	EFS [16]	Proposed Pearson Method
Accuracy (%)	81.5	75	86.5	94.85
Time(milli-seconds)	360	150	550	07

Table V describes the comparison values of accuracy in percentage and time in milli-seconds of proposed Pearson correlation method with top 25% of features existing online feature selection methods.

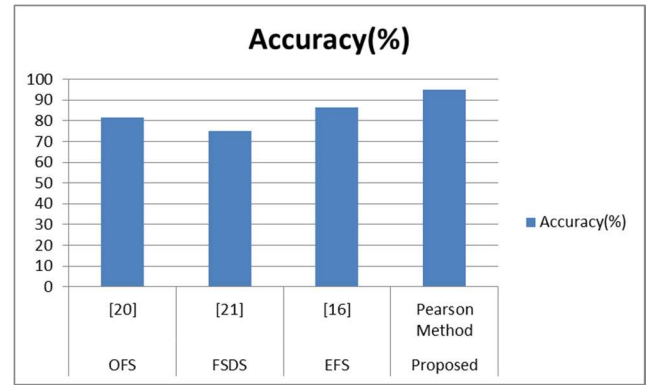


Figure 8: Accuracy (%) comparison of online Pearson feature selection with existing methods

Figure 8 displays a comparison between the proposed online feature selection approach and existing methods using a bar graph. We compared our method's findings to the top 25 percent of existing method's features. In these conditions, the proposed Pearson correlation feature selection method delivers greater accuracy than alternatives.



Figure 9: Time comparison in (milli-seconds) of online Pearson feature selection with existing methods.

Figure 9 displays a comparison between the proposed online feature selection approach and existing methods using a bar graph. We compared our method's findings to the top 25 percent of existing method's features. In these conditions, the proposed Pearson correlation feature selection method takes less time in terms of milli-seconds than alternatives.

VI. CONCLUSION

We have used Pearson correlation method to find the most prominent features for the continuous blocks of data. The accuracy comparison of Pearson correlation is done with complete feature list of HAR dataset and we found that Pearson correlation feature selection method for the given threshold 0.4 gives a good average accuracy of 94.85% compared to complete feature list. In future the proposed work can be compared with other online feature selection methods with filter and wrapper approaches with variety of datasets. Computational challenges can also be addressed in future and comparison with multiple classifiers will be another future scope.

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